

Renal Physiology Overview

The kidney maintains extracellular fluid volume through a set of tightly coupled regulatory mechanisms. Glomerular filtration produces an ultrafiltrate of plasma that is subsequently modified along the nephron.

Tubular reabsorption recovers the majority of filtered sodium. The proximal tubule accounts for roughly two thirds of this reabsorption, and the loop of Henle establishes the medullary osmotic gradient.

